

A Six-Parameter Extension of the Direct Somos–4 Algebraic Generating Function

Thomas Scheuerle

Abstract

The preceding paper constructed a five-parameter quadratic algebraic generating function whose first four Hankel determinants are (λ, m, r, η) and whose Hankel sequence satisfies the normalized Somos–4 recurrence with coefficient τ . It is natural to ask whether the same method extends to the more general two-coefficient recurrence

$$H_{k+4}H_k = \alpha H_{k+3}H_{k+1} + \tau H_{k+2}^2.$$

A weighted transport with $q^6 = \alpha$ gives an exact six-parameter construction, and it preserves the prescribed ordinary Hankel determinants. This selected generating-function branch, however, involves fractional powers of α ; already its first nonconstant coefficient is $q^{-5}r$. Recent work gives broad rational-function-field constructions for general (α, β) –Somos–4 Hankel transforms, so the sharper question here is more constrained: whether one can find a quadratic-algebraic representative over the rational six-parameter field while independently prescribing the four initial determinants and retaining the companion normalization $D_N = H_{N+2}$.

The obstruction is not in the Somos–4 orbit itself, nor in the associated elliptic curve. The Hankel sequence is already defined over the rational six-parameter field, and the transported discriminant admits the rational quartic model

$$Y^2 = (1 - (A + \alpha B)z + \tau z^2)^2 - 4\alpha^2 z^3.$$

We then compare this construction with the generic quadratic-orthogonal-pair realization of Chang and Liu. A direct shift identifies their bilateral orbit with ours, and their subdiagonal data satisfy $d_n = B_{n+2}$. The comparison shows that three notions need not descend together: the Hankel sequence, the algebraic curve, and a chosen moment branch. The first two do descend, while the distinguished transported and Chang–Liu branches do not descend generically. Several natural repairs are ruled out, including parity rescaling, geometric scaling, and injective finite prefixes over the base field. A genuinely non-diagonal construction from the invariant even and odd parts remains possible, so no general nonexistence statement is claimed.

1 Introduction

This paper extends the five-parameter generating function of [3] and clarifies a point that becomes visible only in the six-parameter version. The earlier construction prescribes

$$(H_1, H_2, H_3, H_4) = (\lambda, m, r, \eta)$$

and yields the normalized recurrence

$$H_{k+4}H_k = H_{k+3}H_{k+1} + \tau H_{k+2}^2, \quad k \geq 1. \tag{1}$$

The natural extension is to allow the first bilinear coefficient to vary:

$$H_{k+4}H_k = \alpha H_{k+3}H_{k+1} + \tau H_{k+2}^2, \quad k \geq 1. \tag{2}$$

The six independent parameters are therefore $(\lambda, m, r, \eta, \alpha, \tau)$, and $\alpha = 1$ recovers the normalization of the preceding paper.

Two questions drive the discussion. First, can the earlier algebraic construction be extended to arbitrary values of these six parameters while preserving the prescribed four Hankel determinants? A weighted change of parameters and variable gives an affirmative answer. Second, can one produce a quadratic-algebraic generating function directly over

$$K = \mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau)$$

without fractional powers of α , while keeping the companion identity $D_N = H_{N+2}$? The weighted construction does not settle that second point: it lives naturally over a field containing a sixth root of α , and the first nonconstant coefficient already shows that this dependence is real for the chosen branch.

It is helpful to separate three different objects:

the Hankel sequence, the associated algebraic curve, a chosen generating-function branch.

The first two have direct descriptions over K , while a chosen branch need not. Recent work of Liu, Wang, and Zhang [2] extends the Wang–Zhang sufficient condition over rational-function fields. Our question is narrower because the initial determinants and companion identity are prescribed. This distinction explains why clearing radicals from the curve does not by itself clear them from the normalized generating function.

Sections 2 and 3 construct the six-parameter extension and locate the first fractional power. Section 4 shows that the Hankel orbit and the associated curve nevertheless descend to K . Section 5 compares the result with Chang and Liu’s generic quadratic-algebraic Hankel construction and gives an explicit dictionary between the two descriptions. Section 6 studies the remaining branch and normalization problem and records several transformations that cannot repair it. Section 7 states precisely what remains open.

2 From the five-parameter construction to six parameters

2.1 The five-parameter input

Let

$$G_{1;\lambda,m,r,\eta,\tau}(x) = \sum_{k \geq 0} g_k x^k$$

be the selected algebraic series of [3]. Define

$$H_N(g) = \det(g_{i+j})_{0 \leq i,j \leq N-1}, \quad D_N(g) = \det(g_{i+j+1})_{0 \leq i,j \leq N-1}.$$

The normalization in the preceding paper is

$$(H_1, H_2, H_3, H_4) = (\lambda, m, r, \eta), \quad D_N = H_{N+2}. \quad (3)$$

Its quadratic discriminant factors as

$$\Delta_1(x) = (x\lambda^2\eta mr^2)^2 \left((1 - Jx + \tau x^2)^2 - 4x^3 \right), \quad (4)$$

where

$$J = \frac{\lambda\eta}{mr} + \frac{m^2}{\lambda r} + \frac{r^2}{m\eta} + \frac{\tau mr}{\lambda\eta}. \quad (5)$$

These statements are the only inputs from the five-parameter construction needed below.

2.2 Weighted transport

Work over

$$\mathbb{K} = \mathbb{Q}(\lambda, m, r, \eta, q, \tau), \quad q^6 = \alpha,$$

and introduce the primed parameters

$$\lambda' = \lambda, \quad m' = q^{-2}m, \quad r' = q^{-6}r, \quad \eta' = q^{-12}\eta, \quad \tau' = q^{-8}\tau. \quad (6)$$

Define

$$G_{\alpha; \lambda, m, r, \eta, \tau}(x) := G_{1; \lambda', m', r', \eta', \tau'}(qx). \quad (7)$$

If

$$G_{1; \lambda', m', r', \eta', \tau'}(x) = \sum_{k \geq 0} a_k x^k, \quad G_{\alpha; \lambda, m, r, \eta, \tau}(x) = \sum_{k \geq 0} b_k x^k,$$

then

$$b_k = q^k a_k. \quad (8)$$

Proposition 1 (Determinant weights). *For every $N \geq 1$,*

$$H_N(b) = q^{N(N-1)} H_N(a), \quad D_N(b) = q^{N^2} D_N(a). \quad (9)$$

Proof. In the ordinary Hankel matrix, its (i, j) entry is multiplied by q^{i+j} . Factoring q^i from row i and q^j from column j gives the exponent $2 \sum_{j=0}^{N-1} j = N(N-1)$. For the shifted matrix the entry is multiplied by q^{i+j+1} . The additional factor q in every row contributes N , so the total exponent is $N(N-1) + N = N^2$. $\square$

2.3 The six-parameter Somos-4 recurrence

Theorem 2. *Let $\hat{H}_N = H_N(b)$. Then*

$$(\hat{H}_1, \hat{H}_2, \hat{H}_3, \hat{H}_4) = (\lambda, m, r, \eta)$$

and

$$\hat{H}_{k+4} \hat{H}_k = \alpha \hat{H}_{k+3} \hat{H}_{k+1} + \tau \hat{H}_{k+2}^2, \quad k \geq 1. \quad (10)$$

Proof. Equation (9) and (6) give

$$\hat{H}_1 = \lambda, \quad \hat{H}_2 = q^2 m' = m, \quad \hat{H}_3 = q^6 r' = r, \quad \hat{H}_4 = q^{12} \eta' = \eta.$$

Write $H'_N = H_N(a)$. The primed sequence satisfies

$$H'_{k+4} H'_k = H'_{k+3} H'_{k+1} + \tau' H'_{k+2}^2.$$

The weight difference between the left-hand product and the first product on the right is 6, while that between the left-hand product and H'_{k+2}^2 is 8. Hence

$$\hat{H}_{k+4} \hat{H}_k = q^6 \hat{H}_{k+3} \hat{H}_{k+1} + q^8 \tau' \hat{H}_{k+2}^2.$$

Now use $q^6 = \alpha$ and $q^8 \tau' = \tau$. $\square$

Corollary 3 (Low-order recurrence check). *The first two values forced beyond the initial tuple are*

$$\hat{H}_5 = \frac{\alpha \eta m + \tau r^2}{\lambda}, \quad \hat{H}_6 = \frac{\alpha \hat{H}_5 r + \tau \eta^2}{m}.$$

These formulas provide a direct indexing check for any coefficient expansion of (7).

3 Why fractional powers appear

The weighted transport solves the ordinary Hankel-determinant problem, but it does not preserve the companion normalization in its original form. This is the point at which the stronger rationality question enters.

3.1 The transported companion identity

Proposition 4. *If the primed series obeys $D_N(a) = H'_{N+2}$, then*

$$\hat{D}_N := D_N(b) = q^{-3N-2} \hat{H}_{N+2}, \quad N \geq 1. \quad (11)$$

Proof. By (9),

$$D_N(b) = q^{N^2} H'_{N+2}, \quad \hat{H}_{N+2} = q^{(N+2)(N+1)} H'_{N+2}.$$

Their exponent difference is $N^2 - (N+2)(N+1) = -3N-2$. $\square$

Remark 5. A constant coefficient rescaling $b_k \mapsto cb_k$ changes the ratio $D_N(b)/\hat{H}_{N+2}$ by c^{-2} . It cannot remove the factor q^{-3N} . Thus the literal identity $D_N = H_{N+2}$ is not invariant under this graded transport. Equation (11), rather than the ungraded identity, is the companion normalization naturally inherited from the five-parameter series.

3.2 The first coefficients and failure of descent

For the five-parameter selected series, the companion normalization gives $g_0 = \lambda$ and $g_1 = r$. Applied to the primed parameters in (6), this yields $a_0 = \lambda$ and $a_1 = r' = q^{-6}r$. Therefore (8) gives

$$b_0 = \lambda, \quad b_1 = q^{-5}r = \frac{qr}{\alpha}. \quad (12)$$

The next two coefficients obtained by direct expansion are

$$b_2 = \frac{m}{\lambda} + \frac{q^{-10}r^2}{\lambda} = \frac{m}{\lambda} + \frac{q^2r^2}{\alpha^2\lambda}, \quad (13)$$

$$b_3 = \frac{q^5m^2}{\lambda^2r} + \frac{q^3\eta}{\alpha r} + \frac{2qmr}{\alpha\lambda^2} + \frac{q^3r^3}{\alpha^3\lambda^2}. \quad (14)$$

Hence

$$\begin{aligned} G_\alpha(x) &= \lambda + \frac{qr}{\alpha}x + \left(\frac{m}{\lambda} + \frac{q^2r^2}{\alpha^2\lambda} \right) x^2 \\ &\quad + \left(\frac{q^5m^2}{\lambda^2r} + \frac{q^3\eta}{\alpha r} + \frac{2qmr}{\alpha\lambda^2} + \frac{q^3r^3}{\alpha^3\lambda^2} \right) x^3 + O(x^4). \end{aligned} \quad (15)$$

The first coefficient also follows from (11) at $N = 1$:

$$b_1 = D_1(b) = q^{-5}H_3(b) = q^{-5}r.$$

Thus the fractional power is forced by the transported companion data rather than caused by an insufficient simplification.

Put

$$K = \mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau), \quad L = K(q), \quad q^6 = \alpha.$$

Proposition 6 (Non-descent of the transported branch). *Regard α as an indeterminate. In the generic nondegenerate case $r \neq 0$, the transported series (7) does not belong to $K[[x]]$.*

Proof. Its coefficient of x is $b_1 = q^{-5}r = qr/\alpha$. The polynomial $X^6 - \alpha$ is irreducible over K by Eisenstein's criterion at the prime α in the corresponding polynomial ring. Hence $q \notin K$. Since $r \neq 0$ generically, $b_1 \notin K$, and the transported series is not in $K[[x]]$. $\square$

Although the coefficients do not descend, the fractional powers cancel in the ordinary Hankel determinants. For example,

$$H_2(b) = b_0 b_2 - b_1^2 = \lambda \left(\frac{m}{\lambda} + \frac{q^{-10} r^2}{\lambda} \right) - q^{-10} r^2 = m.$$

Together with the theorem above, this establishes the precise distinction: ordinary Hankel descent holds, whereas descent of the selected transported series fails.

4 What is defined over the six-parameter field

Although the selected transported series does not descend, this does not imply that the underlying recurrence or its curve requires fractional parameters. We now separate these levels.

4.1 Transport of the discriminant

Separate the old invariant into

$$A = \frac{\lambda \eta}{mr} + \frac{\tau mr}{\lambda \eta}, \quad B = \frac{m^2}{\lambda r} + \frac{r^2}{m \eta}. \quad (16)$$

After substitution of (6),

$$J' = q^{-4} A + q^2 B. \quad (17)$$

Consequently the substitution $x \mapsto qx$ gives

$$qJ' = q^{-3} A + q^3 B = \alpha^{-1/2} A + \alpha^{1/2} B. \quad (18)$$

Proposition 7 (Transported discriminant). *The discriminant obtained by substituting (6) and $x \mapsto qx$ into (4) is*

$$\Delta_\alpha(x) = q^{-50} (x \lambda^2 \eta m r^2)^2 \left[\left(1 - (q^{-3} A + q^3 B) x + q^{-6} \tau x^2 \right)^2 - 4 q^3 x^3 \right]. \quad (19)$$

Equivalently,

$$\Delta_\alpha(x) = \alpha^{-25/3} (x \lambda^2 \eta m r^2)^2 \left[\left(1 - (\alpha^{-1/2} A + \alpha^{1/2} B) x + \frac{\tau}{\alpha} x^2 \right)^2 - 4 \alpha^{1/2} x^3 \right]. \quad (20)$$

Proof. The square prefactor has total q -weight $2(1 - 12 - 2 - 12) = -50$. The three terms in the residual factor transform as $J'x \mapsto qJ'x$, $\tau'x^2 \mapsto q^2\tau'x^2 = q^{-6}\tau x^2$, and $x^3 \mapsto q^3x^3$. Equations (17) and $q^6 = \alpha$ give the stated forms. $\square$

The overall monomial factor in (19) is a square in $\mathbb{K}(x)$ and does not affect the residual double cover. It should, however, be retained when comparing discriminants of particular polynomial representatives, because multiplying a quadratic equation by a scalar squares that scalar in its discriminant.

4.2 A rational model of the curve

Remove the square prefactor in (20) and put

$$x = \alpha^{1/2} z. \quad (21)$$

The residual equation becomes

$$Y^2 = \left(1 - (A + \alpha B) z + \tau z^2 \right)^2 - 4 \alpha^2 z^3. \quad (22)$$

All coefficients of (22) belong to $\mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau)$.

For comparison with the quartic convention used in [3], set $X = z^{-1}$ and $y = z^{-2}Y$ (equivalently, $Y = z^2y$). Then (22) is birational to

$$y^2 = \left(X^2 - (A + \alpha B)X + \tau\right)^2 - 4\alpha^2 X. \quad (23)$$

Thus the curve descends to the rational six-parameter field. The coordinate change (21) nevertheless rescales coefficients of a formal power series and does not preserve the normalized Hankel data. Curve descent and descent of the distinguished generating series are therefore different problems.

4.3 Formal representatives with the same Hankel determinants

Preservation of the ordinary Hankel transform alone does not characterize the distinguished algebraic series. For a nondegenerate Hankel sequence define

$$\beta_n = \frac{H_{n-1}H_{n+1}}{H_n^2}, \quad H_0 = 1. \quad (24)$$

All H_n generated by (10), and hence all β_n , lie in K . For arbitrary $a_0, a_1, \dots \in K$, the Jacobi fraction

$$\tilde{G}(x) = \frac{H_1}{1 - a_0x - \frac{\beta_1 x^2}{1 - a_1x - \frac{\beta_2 x^2}{1 - a_2x - \ddots}}} \quad (25)$$

has ordinary Hankel determinants H_n . Thus radical-free formal-series representatives over K exist, but they need not be quadratic algebraic, preserve the graded companion identity, or arise from the distinguished Stieltjes splitting.

5 Comparison with the construction of Chang and Liu

The preceding results leave open whether a different quadratic-algebraic generating function can avoid fractional powers. The construction of Chang and Liu is directly relevant because it realizes a generic bilateral Somos–4 orbit by Hankel determinants. We therefore compare the two descriptions explicitly before considering further repairs.

5.1 The shifted Somos–4 orbit

Chang and Liu [1, Sections 3.3–4.2] treat the generic bilateral Somos–4 initial-value problem by Hankel determinants of quadratic orthogonal pairs. Their normalized recurrence is

$$\sigma_{n+2}\sigma_{n-2} = a\sigma_{n+1}\sigma_{n-1} + b\sigma_n^2, \quad n \in \mathbb{Z}, \quad (26)$$

with $\sigma_{-1} = x$, $\sigma_0 = 1$, $\sigma_1 = y$, and $\sigma_2 = z$. Our freely prescribed determinants fit this normalization after the shift

$$\sigma_n = \frac{H_{n+2}}{m}, \quad (x, y, z; a, b) = \left(\frac{\lambda}{m}, \frac{r}{m}, \frac{\eta}{m}; \alpha, \tau\right). \quad (27)$$

This uses all six parameters and avoids the constraint $\eta = \alpha\lambda r + \tau m^2$ that would follow from imposing our recurrence at $N = 0$ with $H_0 = 1$.

Write $u^2 = \alpha$ and define

$$\mathcal{N} = \tau r^2 m + \lambda^2 \eta^2 + \alpha \eta m^3 + \alpha \lambda r^3, \quad (28)$$

$$\mathcal{N}_0 = \tau r^2 m - \lambda^2 \eta^2 + \alpha \eta m^3 + \alpha \lambda r^3. \quad (29)$$

Under (27), the Chang–Liu parameters include

$$d_0 = \frac{\lambda r}{m^2}, \quad u_0 = \frac{2r}{m}, \quad (30)$$

$$\frac{v}{u} = \frac{\mathcal{N}}{2\alpha\lambda r\eta m}, \quad uv = \frac{\mathcal{N}}{2\lambda r\eta m}, \quad (31)$$

$$\frac{v_0}{u} = \frac{\mathcal{N}_0}{2\alpha\lambda r\eta m}, \quad uv_0 = \frac{\mathcal{N}_0}{2\lambda r\eta m}, \quad (32)$$

and

$$f = \frac{\tau}{\alpha} - \frac{\mathcal{N}^2}{4\alpha\lambda^2 r^2 \eta^2 m^2}. \quad (33)$$

The sign difference between $\mathcal{N}$ and $\mathcal{N}_0$ is essential because v_0 enters the distinguished moment branch. The scaling $X = uZ$ turns their curve into

$$Y^2 = (\alpha Z^2 + f)^2 + 4\alpha Z - 4uv, \quad (34)$$

which is defined over K . This is curve-level descent, not coefficientwise descent of a selected moment branch.

5.2 Translation to Jacobi data

Use the local Jacobi convention

$$F(t) = \frac{\lambda}{1 - A_0 t - \frac{B_1 t^2}{1 - A_1 t - \frac{B_2 t^2}{1 - A_2 t - \frac{B_3 t^2}{1 - A_3 t - B_4 t^2 M_4(t)}}}}, \quad M_4(0) = 1. \quad (35)$$

The product formula

$$H_N = \lambda^N \prod_{j=1}^{N-1} B_j^{N-j} \quad (36)$$

gives

$$B_1 = \frac{m}{\lambda^2}, \quad B_2 = \frac{\lambda r}{m^2}, \quad B_3 = \frac{m\eta}{r^2}, \quad B_4 = \frac{r(\alpha\eta m + \tau r^2)}{\lambda\eta^2}. \quad (37)$$

The companion identities through $D_3 = H_5$ force

$$A_0 = \frac{r}{\lambda}, \quad A_1 = \frac{\eta\lambda^2 + m^2}{\lambda m r}, \quad A_2 = \frac{\alpha\eta m^3 + \lambda r^3 + \tau m^2 r^2}{\eta\lambda m r}. \quad (38)$$

Proposition 8 (Fourth shifted determinant). *For the convention (35),*

$$D_4 = \lambda^4 B_1^3 B_2^2 B_3 (A_0 A_1 A_2 A_3 - A_0 A_1 B_3 - A_0 A_3 B_2 - A_2 A_3 B_1 + B_1 B_3). \quad (39)$$

After substitution of (37) and (38), this reduces to

$$D_4 = \frac{\eta}{\lambda r^2} [A_3 r(\alpha\eta m + \tau r^2) - \eta^2 \lambda]. \quad (40)$$

Consequently $D_4 = H_6$, where

$$H_6 = \frac{\alpha^2 \eta m r + \alpha \tau r^3 + \tau \eta^2 \lambda}{\lambda m},$$

uniquely forces

$$A_3 = \frac{\alpha^2 \eta m r^3 + \alpha \tau r^5 + \eta^3 \lambda m + \tau \eta^2 \lambda r^2}{\eta m r (\alpha \eta m + \tau r^2)}. \quad (41)$$

Proof. Equation (39) follows by direct expansion of the shifted 4×4 Hankel determinant in Jacobi parameters. Substitution and collection of the terms linear in A_3 gives (40). Solving (40) for A_3 after inserting the displayed value of H_6 gives (41). $\square$

An equivalent useful form is

$$A_3 = \frac{\alpha r^2}{\eta m} + \frac{\eta \lambda (\eta m + \tau r^2)}{m r (\alpha \eta m + \tau r^2)}.$$

No B_4 occurs in D_4 : the fourth ordinary Hankel step determines B_4 , while the fourth shifted step determines A_3 , without imposing any extra relation among the six parameters.

Chang and Liu use

$$d_n = \frac{\sigma_{n-1} \sigma_{n+1}}{\sigma_n^2}.$$

Substitution of (27) gives the exact all-order interface

$$d_n = \frac{H_{n+1} H_{n+3}}{H_{n+2}^2} = B_{n+2}. \quad (42)$$

Thus $d_0 = B_2$, $d_1 = B_3$, and $d_2 = B_4$. At the subdiagonal level only B_1 precedes the Chang–Liu tail; the diagonal and moment normalizations remain separate issues.

6 What the comparison resolves

The comparison confirms the same basic distinction found for the weighted transport: the orbit and the curve may descend even when one distinguished moment branch does not.

6.1 Even and odd parts of the Chang–Liu branch

Let

$$S(t; u) = \sum_{n \geq 0} s_n(u) t^n, \quad u^2 = \alpha,$$

be the distinguished Chang–Liu moment branch. Chang and Liu, Remark 4.7, exhibit the two square-root choices and the alternating-sign relation

$$S(t; -u) = S(-t; u), \quad s_n(-u) = (-1)^n s_n(u). \quad (43)$$

Hence the two branches have equal ordinary Hankel determinants, but a single branch is generically not fixed by $u \mapsto -u$ and therefore does not descend coefficientwise to $K[[t]]$. Writing

$$S(t; u) = E(t^2; u) + t O(t^2; u)$$

gives $E(z; u) \in K[[z]]$ and $O(z; u)/u \in K[[z]]$, whereas O is generically anti-invariant.

Proposition 9 (Parity-rescaling obstruction). *Let $r_{2k} = a s_{2k}$ and $r_{2k+1} = b s_{2k+1}$. If the generic identity $H_3(r) = \kappa H_3(s)$ holds coefficientwise, then $a^3 = \kappa$, $ab^2 = \kappa$, and therefore $b^2 = a^2$.*

Proof. Compare the coefficients of the independent monomials in the two generic 3×3 Hankel determinants. The pure even contribution has weight a^3 , while the terms containing two odd moments have weight ab^2 . $\square$

In particular, the descended recombination $R(t) = E(t^2; u) + tO(t^2; u)/u$ fails already at order two:

$$H_2(R) - H_2(S) = \left(1 - \frac{1}{\alpha}\right) s_1^2.$$

Proposition 10 (Geometric-scaling obstruction). *No generic scaling $r_n = cd^n s_n$ converts the normalized Chang–Liu determinants $H_N(s) = H_{N+2}/m$ into $H_N(r) = H_{N+2}$ for all N .*

Proof. Such a scaling gives $H_N(r) = c^N d^{N(N-1)} H_N(s)$, so one would need $c^N d^{N(N-1)} = m$. The cases $N = 1, 2$ force $c = m$ and $d^2 = m^{-1}$, while $N = 3$ then gives $c^3 d^6 = 1$, not m , generically. $\square$

A finite Jacobi prefix defined over K , when its denominator is invertible, is an injective K -rational linear-fractional transformation of its tail. The Galois-equivariant obstruction below therefore shows that it cannot turn a non-descending Chang–Liu branch into a descending one. This excludes the direct prefix strategy, but not every possible quadratic-algebraic representative.

6.2 Why base-field transforms cannot repair a branch

The following observation excludes a broad class of attempted rationalizations.

Proposition 11 (Galois-equivariant obstruction). *Let $T : L[[x]] \rightarrow L[[x]]$ be injective and commute with every K -automorphism of L . If $T(G) \in K[[x]]$, then $G \in K[[x]]$. If T is invertible, the two conditions are equivalent.*

Proof. For every $\sigma \in \text{Gal}(M/K)$, $T(\sigma G) = \sigma T(G) = T(G)$. Injectivity gives $\sigma G = G$, so G is fixed by the Galois action and belongs to $K[[x]]$. $\square$

For $c \in K$, the binomial transform

$$B_c(G)(x) = \frac{1}{1 - cx} G\left(\frac{x}{1 - cx}\right)$$

is defined over K , commutes with the Galois action, and is invertible with inverse B_{-c} . Likewise, the invert transform

$$I_c(G)(x) = \frac{G(x)}{1 - cxG(x)}$$

is defined over K , Galois-equivariant, and invertible with inverse I_{-c} . Consequently neither transform can turn the non-descending transported branch into a series in $K[[x]]$. This conclusion concerns these transforms over the base field; it does not exclude a new algebraic construction with different companion data.

7 Remaining possibilities

The established facts now separate three levels cleanly: the Chang–Liu curve descends; its ordinary Hankel sequence descends; an individual distinguished moment branch does not descend generically. The rational Jacobi calculation extends through A_3 and B_4 , and the subdiagonal interface is exactly $d_n = B_{n+2}$.

The remaining constructive possibility should therefore be sought in the invariant pair

$$E(z), \quad R(z) := O(z)/u,$$

rather than by attaching an injective K -rational prefix to one raw branch. Starting from the quadratic equation for $S(t; u)$, one should substitute

$$S(t; u) = E(t^2) + utR(t^2)$$

and separate invariant and anti-invariant parts. This yields coupled algebraic equations over K for E and R . A non-diagonal rational construction from this pair can then be tested against

$$f_0 = \lambda, \quad f_1 = r, \quad f_2 = \frac{r^2 + m}{\lambda}$$

and the determinant conditions through $D_4 = H_6$.

Thus no general nonexistence claim is made. What is excluded is the narrower class consisting of parity rescalings, constant mass and argument scalings, alternating signs, and injective finite prefixes over K applied to a single distinguished Chang–Liu branch. A constructive all-order theorem, or a precise nonexistence theorem for a broader class, remains open.

8 Conclusion

The starting point was the five-parameter generating function of [3]. The weighted substitution (7) extends it to the full six-parameter recurrence, and it does so exactly: it preserves the prescribed ordinary Hankel determinants and yields the recurrence (10). At the level of the orbit itself, the six-parameter Hankel problem is therefore solved.

The same construction also exposes a second, more delicate issue. Its companion determinants satisfy the graded identity (11), and the coefficient $b_1 = q^{-5}r$ shows that the selected branch does not generically descend to $K = \mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau)$. This is not a mere artifact of an unsimplified formula. By contrast, the rational quartic (22) shows that the associated curve does descend. Thus the Hankel sequence, the curve, and a distinguished generating-function branch must be treated as separate objects.

The comparison with Chang and Liu makes the same point from a second angle. The shifted dictionary (27) identifies their generic bilateral orbit with ours, and the all-order relation $d_n = B_{n+2}$ identifies the subdiagonal Jacobi data. The rational restoration through A_3 and B_4 confirms that the prescribed companion data are consistent through the computed levels. Nevertheless, the parity relation (43) shows that an individual distinguished moment branch is generically non-descending.

The remaining question is therefore narrower than the original one. It is unknown whether another quadratic-algebraic representative can simultaneously have coefficients in K , realize the prescribed six-parameter Hankel orbit, and satisfy the ungraded identity $D_N = H_{N+2}$. Parity rescaling, geometric scaling, and injective finite prefixes over K cannot achieve all three goals. These obstructions do not constitute a general nonexistence theorem. A non-diagonal construction based on the invariant pair $(E, O/u)$ remains a concrete possibility for further investigation.

Disclosure on authoring and verification tools

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References

- [1] X.-K. Chang and J. Liu, Hankel determinants from quadratic orthogonal pairs for hyperelliptic functions and their applications, arXiv:2603.11670v4 [nlin.SI], 2026; to appear in the *Journal of the London Mathematical Society*.
- [2] F. Liu, Y. Wang, and Z. Zhang, Hankel transform and (α, β) Somos–4 sequences, arXiv:2608.08703v1 [math.CO], 2026.
- [3] T. Scheuerle, Direct generation of a Somos–4 sequence from an algebraic generating function, arXiv:2609.15754 [math.CO], 2026.
- [4] H. S. Wall, *Analytic Theory of Continued Fractions*, Van Nostrand, New York, 1948.
- [5] P. Flajolet, Combinatorial aspects of continued fractions, *Discrete Mathematics* **32** (1980), 125–161.
- [6] J. W. Layman, The Hankel transform and some of its properties, *Journal of Integer Sequences* **4** (2001), Article 01.1.5.
- [7] C. French, Transformations preserving the Hankel transform, *Journal of Integer Sequences* **10** (2007), Article 07.7.3.
- [8] A. N. W. Hone, Elliptic curves and quadratic recurrence sequences, *Bulletin of the London Mathematical Society* **37** (2005), 161–171.
- [9] S. Fomin and A. Zelevinsky, The Laurent phenomenon, *Advances in Applied Mathematics* **28** (2002), 119–144.
- [10] A. J. van der Poorten and C. S. Swart, Recurrence relations for elliptic sequences: every Somos–4 is a Somos– k , *Bulletin of the London Mathematical Society* **38** (2006), 546–554.